<u>Math-2201(Practice Problem)</u>

Lecture no		Chapter	Topics
	Rectangular coordinates in 3-space, dot and cross products of vectors, parametric equation of straight lines.	11.1	Lecture
		11.2	Example-1, 2, 3, 4, 5, 6, 7 Exercise-5,6,9,21,24,32,29(a),40,57
		11.3	Example-1, 2, 3, 4, 5, 7 Exercise-1,2,3,15,24,25,26,27
		11.4	Example-1, 2, 3, 4, 5, Exercise-3-6,10,11,18,20,23,24,25,26,27 ,29(a),34
		11.5	Example-1,2,3,4,5,6 Exercise set-11.5 (3, 5, 7, 9, 15, 16, 19, 21, 23(a, b), 25, 29, 31, 33, 35, 37)
	Plane in 3-space, quadratic surfaces	11.6	Example-1, 2, 3, 4, 5, 6, 7, 8, 9, 10 Exercise set-11.6(1,3, 5, 7, 9, 11, 13, 15, 17, 19, 25, 27, 29, 31, 33, 35, 37, 41, 43, 45, 47)
	Double and triple integral in rectangular coordinate systems.	14.1 14.2 14.5	**14.1:** Theorem 14.1.3, Examples: 1-4, Exercise Set 14.1: 1-16 **14.2:** Definition 14.2.1, Theorem 14.2.2, Examples: 1-8 Exercise Set 14.2: 1-8 (odd numbers), 9,10,15-18 **14.5:** Theorem 14.5.1-14.5.2, Examples: 1-5 Exercise Set 14.5: 1-12
	Triple integral in rectangular, polar, cylindrical and spherical coordinate systems.	11.8 14.6	Exercise- set 11.8(1,3,5,7,9,11) 14.6: Example-1,2,3,4
	Differentiation and integration of vector-valued function, directional derivative, and gradient of scalar fields.	12.1	Example-1,2
		12.2	Example-1,2,3,7,8,9
		13.6	Exercise set-13.6(9,11,13,53,55,61,63)
	Vector fields, line integrals, conservative vector field, Green's theorem.	15.1	Example-2 – 5, Exercise-15.1(17, 19, 37, 38)
		15.2	Example- 5, 6, 7 Exercise set 15.2-(7, 9, 13, 23, 25, 27, 33, 34, 37)
		15.3	Example-1,2,3,4,5,6 Exercise set 15.3(1, 3, 5, 9, 11, 13)
		15.4	Example-1, 2, 3 Exercise set 15.4 (1, 3, 5,7, 9,11)
	Surface integral, flux, divergence theorem.	15.5	Example-2, 3 Exercise set- 15.5 (1, 3, 5, 7)
		15.6	Example-2 Exercise set-15.6 (9,11, 13, 15)
		15.7	Example-1,2,3,4 Exercise set-15.7(1,3,9,11, 13, 15, 17)
	Conic sections, rotation of axes	10.4	Example-1, 2, 3, 4, 5, 6, 7, 8, 9, 10
		10.5	Example-1, 2, 3, 4